Database Management Systems Project (Phase 1)

Project Title: Azkar Management system

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1. Introduction

This documentation explains the database design for an Islamic Azkar mobile application.
The goal of the database is to store different types of azkar, track user activity, support multiple languages, allow custom azkar, and manage user accounts and permissions.
The ER diagram shows how all entities in the system are connected and how data flows between them.

2. System Overview

The Azkar application allows users to read, organize, and track their daily azkar in a simple and personalized way.
Users can browse azkar by category, view translations, and adjust settings to match their preferred language and style.
The system records each user's progress, generates daily summaries, and lets users create custom azkar or mark favorites for quick access.
The application also includes security features such as authentication logs ensuring safe access and safety checks.

3. Entities

3.1 User

In our system, User is an entity which represents people who use the application. Each User has a:

- Unique User ID
- Email address
- Username
- Password
- Name, composed of:
  - First Name
  - Last Name

3.2 Role

The Role entity represents the different access levels available in the application. The attributes of the entity are:

- Unique Role ID
- Role Name

3.3 Authentication Log

The Auth Log entity records all log in attempts done by User. Each record stores details such as:

- Unique log ID
- Status of Attempt
- Time of Attempt
- Reason for Failure
- IP Address

3.4 Daily Summary

This weak entity represents an overview of a user's zikr activity in a day, and it depends on the user's entity. For each record, it stores:

- Time spent
- Total Zikr Completed
- Partially unique Summary Date

3.5 Zikr

The Zikr Entity is Represented to aggregate all essential data points related to the remembrance, such as the text, benefits, and references. This entity includes attributes such as:

- Unique Zikr Id
- Virtue text
- Reptation count
- Arabic text
- Hadith Reference
- Audio file

3.6 Settings

This entity is dedicated to storing and managing user-specific configurations for the application. It controls the app's appearance and behavior, covering display options (like themes and fonts) and user interaction (like notifications and vibration). This entity includes attributes such as:

- Unique Settings Id
- Notification time
- Notification sound
- Vibration enabled
- Themes
- Font color
- Font size

3.7 Review

The REVIEW entity represents a rating or opinion submitted by a user about a specific product or service. Its purpose is to store both the numerical rating value and the textual comment, along with the date and time of submission. This entity includes attributes such as:

- Unique Review Id
- Rating value
- Rating comment
- Rating date
    - Day
    - Month
    - Year

3.8 Language

This entity represents a specific language supported within the application or system, primarily to facilitate localization and translation. This entity includes attributes such as:

- Unique Language Id
- Language Name
- IS_RTL
- Language code

3.9 Category

This entity represents a group or section used to organize and cluster similar zikr items within the system. This entity includes attributes such as:

- Unique Category Id
- Category Name
- Category order
- Category Description

4 Relationships

4.1 Access Level

Multiple users must *access level* for several Roles, and each role can be associated with many users.

This relationship allows the User to choose their role either as a normal user or an admin user.

It's attribute:

- Assignment date: The date on which the Role was assigned to the User

Every User *must* be assigned at least one Role.

A Role may belong to Users.

A Role *can* be defined in the system without being assigned to any User yet**.**

4.2 Attempts

Each User *attempts* several Authentication Logs.

This relationship helps the system keep track of login attempts and supports security checks.

Every User *must* be associated with at least one AUTH_LOG record (The system does not allow a user to exist without having logged an attempt).

Every AUTH_LOG record *must* be associated with at least one User who performed the attempt.

4.3 Generates

Each User can *generate* many summaries over time, while every Daily Summary belongs to only one User.

This relationship allows the app to show statistics and streaks.

Every User must generate at least one Daily Summary.

Every Daily Summary must be generated by exactly one User.

4.4 Uses

One Language can be *used by* multiple Settings.

Many different users can choose the same language, but each settings record can be linked to only one language at a time.

Every Settings record must reference an existing Language.

A Language may exist in the system without necessarily being used by any Settings to record.

4.5 Rating

The *Rating* relationship is a quaternary relationship that links User, Zikr, Language, and Review to represent a user's evaluation of a zikr in a specific language, along with any associated written review. The participating entities:

- User
  Represents the user who gives the rating.
- Language
  Represent the Language in which the Zikr is displayed or stored.
- Review
  Represents the textual comment or feedback associated with the rating (if applicable).
- Zikr

Represents the specific zikr being rated.

4.6 Personalizes

Each User must *personalize* only one Settings in configuration, permitting a User to customize their Azkar application to their preferences.

Every User must have a corresponding Settings configuration.

Every Settings configuration can belong to one User.

4.7 Favour

Multiple zikrs can be *favored by* multiple users. It keeps track of:

- Data added: The date when the User added the Zikr to Favorite.

A User can exist without having any favorite Zikrs.

A Custom Zikr may not have been added to any user's favorites.

4.8 Recites

Many Users *recites* multiple Zikrs. It keeps track of:

- Zikr Counter: The number of times the User has recited the specific Zikr

A User may or may not recite any Zikr, and vice versa.

4.9 Log progress

Many users *log progress* of Several zikrs. It keeps track of:

- Actual count: The number of times the User has recited the Zikr
- Completion date (composed of Day, Month, Year): The date when the User completed the Zikr

A User may exist in the system without having any recorded progress in any Zikr.

A Zikr may exist even if no User has recorded progress for it.

4.10 Category translated

Multiple Categories can be *category translated* into many Languages. It keeps track of:

- Category Name: The name of the Category in the specified Language.
- Category Order: The display order of the Category in the system or interface.
- Category Description: A descriptive explanation of the Category in the given Language.

A Category may or may not have a translation in a given Language.

A Language may exist in the system without containing any translated Categories.

4.11 Classifies

This relationship represents that Zikr is *classified* as Categories according to category name.

A Category can *classify* multiple Zikr entries. Each Zikr entry belongs to multiple Categories.

Every Category must classify at least one Zikr.

Every Zikr must belong to at least one Category.

4.12 Translated

This relationship represents that Zikr entries can be *translated into* many languages using these attributes:

- Zikr Text: The translated text of the Zikr in the specified Language
- Virtue Text: A description of the virtues or benefits associated with the Zikr.
- Explanation Text: An explanation or commentary for the Zikr in the given Language.

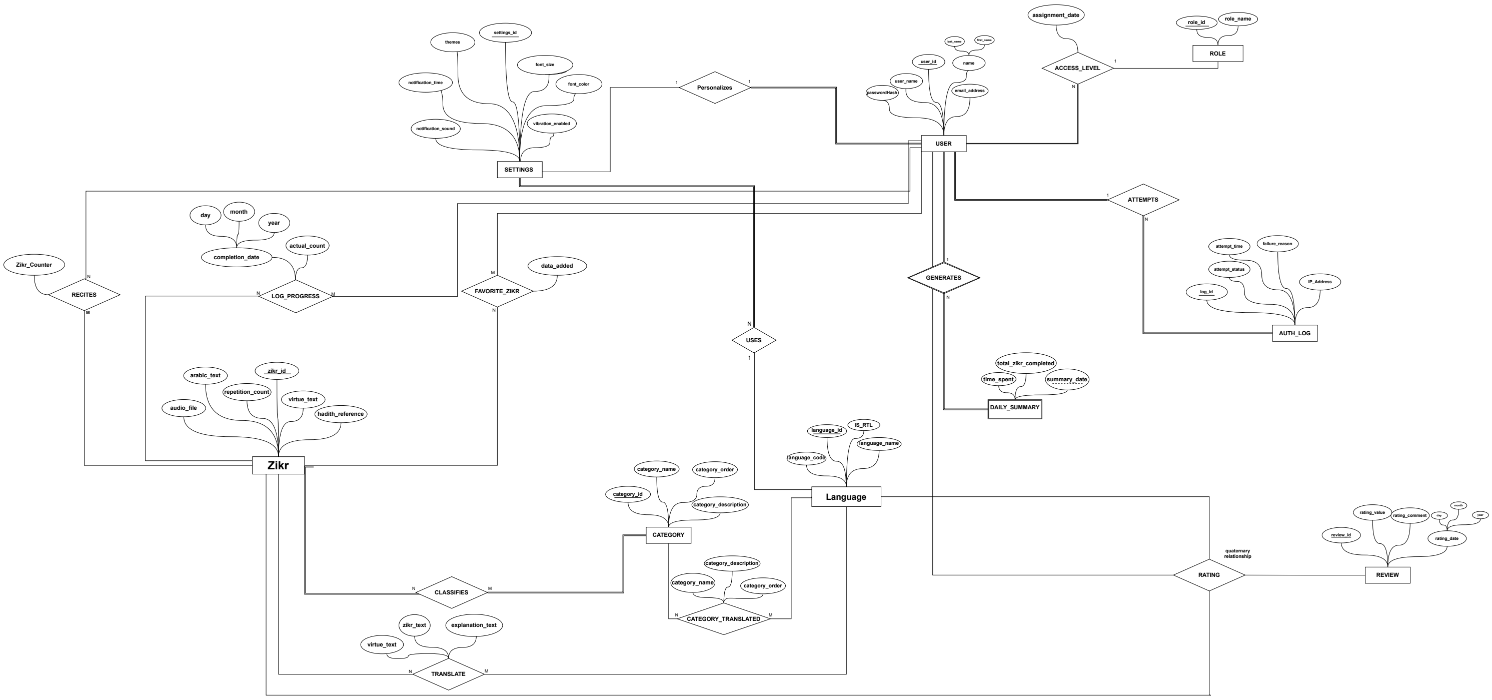
A single Zikr entity can be translated into multiple languages entities. A single Language entity can be translated into multiple distinct Category entities.

A Zikr may exist without being translated into any Language.

A Language may exist in the system without containing any translated Zikrs.

5 Conclusion

Overall, the system is designed to make the app easy to use, personalized, and reliable. It provides a complete structure for reading azkar, tracking progress, supporting multiple languages, and maintaining secure and consistent user experience.



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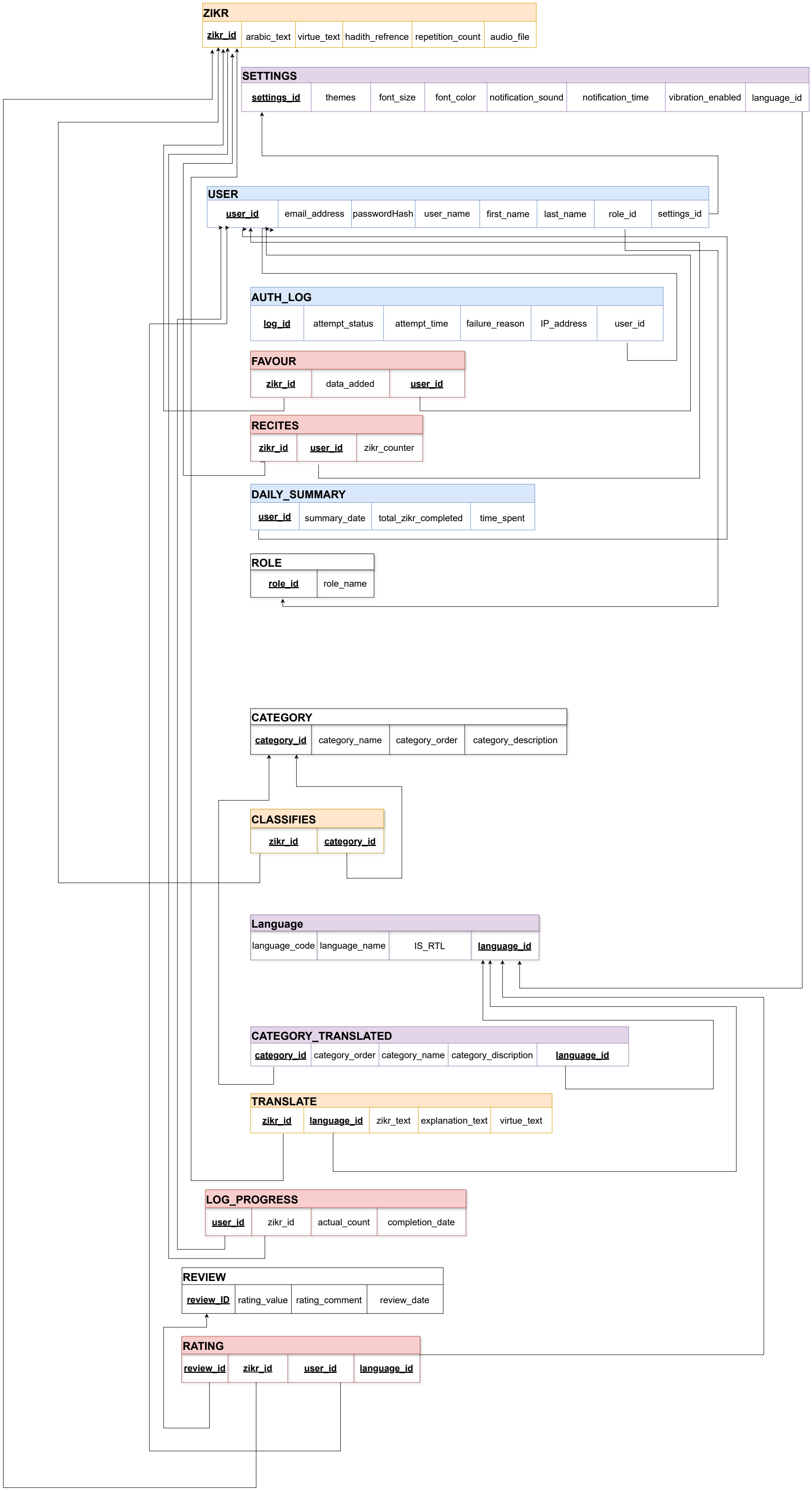
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